# TCP SYN Flood Attack Investigation

**Incident Report**


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## 1. Executive Summary


On July 2026, the web server of a travel agency experienced a service disruption caused by a TCP SYN Flood (Denial-of-Service) attack.

The attack generated a high volume of TCP SYN packets that exhausted server resources and prevented legitimate users from accessing the company's website.

The incident was analyzed using Wireshark packet captures.


## 2. Incident Description

The organization received an alert indicating abnormal web server behavior.

Employees reported connection timeout errors while accessing the vacation sales website.

A packet capture revealed an excessive number of TCP SYN packets originating from the IP address:

203.0.113.0

directed to:

192.0.2.1

## 3. Technical Analysis

During the attack, the attacker continuously sent SYN packets without completing the handshake.

This caused the server to maintain thousands of half-open TCP connections.

## 4. Evidence

## Evidence 1:

| No. | Time | Source | Destination | Protocol | Info |
|---|---|---|---|---|---|
| 47 | 3.144521 | 198.51.100.23 | 192.0.2.1 | TCP | 42584->443 [SYN] Seq=0 Win-5792 Len=120... |
| 48 | 3.195755 | 192.0.2.1 | 198.51.100.23 | TCP | 443->42584 [SYN, ACK] Seq=0 Win-5792 Len=120... |
| 49 | 3.246989 | 198.51.100.23 | 192.0.2.1 | TCP | 42584->443 [ACK] Seq=1 Win-5792 Len=120... |
| 50 | 3.298223 | 198.51.100.23 | 192.0.2.1 | HTTP | GET /sales.html HTTP/1.1 |
| 51 | 3.349457 | 192.0.2.1 | 198.51.100.23 | HTTP | HTTP/1.1 200 OK (text/html) |
| 52 | 3.390692 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 53 | 3.441926 | 192.0.2.1 | 203.0.113.0 | TCP | 443->54770 [SYN, ACK] Seq=0 Win-5792 Len=120... |
| 54 | 3.49316 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [ACK Seq=1 Win=5792 Len=0... |
| 55 | 3.544394 | 198.51.100.14 | 192.0.2.1 | TCP | 14785->443 [SYN] Seq=0 Win-5792 Len=120... |
| 56 | 3.599628 | 192.0.2.1 | 198.51.100.14 | TCP | 443->14785 [SYN, ACK] Seq=0 Win-5792 Len=120... |
| 57 | 3.664863 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 58 | 3.730097 | 198.51.100.14 | 192.0.2.1 | TCP | 14785->443 [ACK] Seq=1 Win-5792 Len=120... |
| 59 | 3.795332 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win-5792 Len=120... |
| 60 | 3.860567 | 198.51.100.14 | 192.0.2.1 | HTTP | GET /sales.html HTTP/1.1 |
| 61 | 3.939499 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win-5792 Len=120... |
| 62 | 4.018431 | 192.0.2.1 | 198.51.100.14 | HTTP | HTTP/1.1 200 OK (text/html) |
| 63 | 4.097363 | 198.51.100.5 | 192.0.2.1 | TCP | 33638->443 [SYN] Seq=0 Win-5792 Len=120... |
| 64 | 4.176295 | 192.0.2.1 | 203.0.113.0 | TCP | 443->54770 [SYN, ACK] Seq=0 Win-5792 Len=120... |
| 65 | 4.255227 | 192.0.2.1 | 198.51.100.5 | TCP | 443->33638 [SYN, ACK] Seq=0 Win-5792 Len=120... |
| 66 | 4.256159 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 67 | 5.235091 | 198.51.100.5 | 192.0.2.1 | TCP | 33638->443 [ACK] Seq=1 Win-5792 Len=120... |
| 68 | 5.236023 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 69 | 5.236955 | 198.51.100.16 | 192.0.2.1 | TCP | 32641->443 [SYN] Seq=0 Win-5792 Len=120... |
| 70 | 5.237887 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 71 | 6.228728 | 198.51.100.5 | 192.0.2.1 | HTTP | GET /sales.html HTTP/1.1 |
| 72 | 6.229638 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 73 | 6.230548 | 192.0.2.1 | 198.51.100.16 | TCP | 443->32641 [RST, ACK] Seq=0 Win-5792 Len=120... |
| 74 | 6.330539 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 75 | 6.330885 | 198.51.100.7 | 192.0.2.1 | TCP | 42584->443 [SYN] Seq=0 Win=5792 Len=0... |
| 76 | 6.331231 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 77 | 7.330577 | 192.0.2.1 | 198.51.100.5 | TCP | HTTP/1.1 504 Gateway Time-out (text/html) |
| 78 | 7.351323 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 79 | 7.360768 | 198.51.100.22 | 192.0.2.1 | TCP | 6345->443 [SYN] Seq=0 Win=5792 Len=0... |
| 80 | 7.380773 | 192.0.2.1 | 198.51.100.7 | TCP | 443->42584 [RST, ACK] Seq=1 Win-5792 Len=120... |
| 81 | 7.380878 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |

## Description:

Large number of SYN packets sent from 203.0.113.0.

# Evidence 2

| | | | | | |
|---|---|---|---|---|---|
| 84 | 7.581629 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 85 | 7.680504 | 192.0.2.1 | 198.51.100.22 | TCP | 443->6345 [RST, ACK] Seq=1 Win=5792 Len=0... |
| 86 | 7.709377 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 87 | 7.738241 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 88 | 7.767105 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 89 | 13.895969 | 192.0.2.1 | 203.0.113.0 | TCP | 443->54770 [RST, ACK] Seq=1 Win=5792 Len=0... |
| 90 | 13.919832 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 91 | 13.943695 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 92 | 13.967558 | 192.0.2.1 | 198.51.100.16 | TCP | 443->32641 [RST, ACK] Seq=1 Win-5792 Len=120... |
| 93 | 13.991421 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 94 | 14.015245 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 95 | 14.439072 | 192.0.2.1 | 203.0.113.0 | TCP | 443->54770 [RST, ACK] Seq=1 Win=5792 Len=0... |
| 96 | 14.862899 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 97 | 14.886727 | 198.51.100.9 | 192.0.2.1 | TCP | 4631->443 [SYN] Seq=0 Win=5792 Len=0... |
| 98 | 15.310554 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 99 | 15.734381 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 100 | 16.158208 | 192.0.2.1 | 203.0.113.0 | TCP | 443->54770 [RST, ACK] Seq=1 Win=5792 Len=0... |
| 101 | 16.582035 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 102 | 17.005862 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 103 | 17.429678 | 192.0.2.1 | 203.0.113.0 | TCP | 443->54770 [RST, ACK] Seq=1 Win=5792 Len=0... |
| 104 | 17.452693 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 105 | 17.475708 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 106 | 17.498723 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 107 | 17.521738 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 108 | 17.544753 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 109 | 17.567768 | 192.0.2.1 | 203.0.113.0 | TCP | 443->54770 [RST, ACK] Seq=1 Win=5792 Len=0... |
| 110 | 17.590783 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 111 | 18.413795 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 112 | 18.436807 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 113 | 18.459819 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 114 | 18.482831 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 115 | 18.506655 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 116 | 18.529667 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 117 | 18.552679 | 192.0.2.1 | 203.0.113.0 | TCP | 443->54770 [RST, ACK] Seq=1 Win=5792 Len=0... |
| 118 | 18.875692 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 119 | 19.198705 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |

## Description:

Repeated SYN packets targeting TCP port 443.

## Evidence 3

| | | | | | |
|---|---|---|---|---|---|
| 127 | 21.782809 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 128 | 22.105822 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 129 | 22.428835 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 130 | 22.751848 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 131 | 23.074861 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 132 | 23.397874 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 133 | 23.720887 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 134 | 24.0439 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 135 | 24.366913 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 136 | 24.689926 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 137 | 25.012939 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 138 | 25.335952 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 139 | 25.658965 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 140 | 25.981978 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 141 | 26.304991 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 142 | 26.628004 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 143 | 26.951017 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 144 | 27.27403 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 145 | 27.597043 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 146 | 27.920056 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 147 | 28.243069 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 148 | 28.566082 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 149 | 28.889095 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 150 | 29.212108 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 151 | 29.535121 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 152 | 29.858134 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 153 | 30.181147 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 154 | 30.50416 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 155 | 30.827173 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 156 | 31.150186 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 157 | 31.473199 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 158 | 31.796212 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 159 | 32.119225 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 160 | 32.442238 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 161 | 32.765251 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |
| 162 | 33.088264 | 203.0.113.0 | 192.0.2.1 | TCP | 54770->443 [SYN] Seq=0 Win=5792 Len=0... |

## Description:

Legitimate HTTP traffic begins failing due to resource exhaustion.

## 5. Root Cause

The attacker exploited the TCP connection establishment process by generating numerous incomplete TCP connection requests.

Server resources became exhausted due to the accumulation of half-open connections.

## 6. Impact

Business impact included:

- Website unavailable.
- Employees unable to access vacation packages.
- Customer requests interrupted.
- Reduced service availability.

## 7. Mitigation Performed

Immediate actions included:

- Temporarily disconnecting the web server.
- Blocking the attacker's IP address.
- Preserving packet captures for investigation.
- Escalating the incident.

## 8. Recommendations

| Priority | Recommendation | Purpose |
|---|---|---|
| High | Enable SYN Cookies | Prevent the server from allocating resources until the TCP three-way handshake is successfully completed, reducing the impact of TCP SYN Flood attacks. |
| High | Implement Rate Limiting | Restrict the number of connection attempts or requests allowed from a single IP address within a defined time period to reduce malicious traffic. |
| Medium | Deploy an Intrusion Detection/Prevention System (IDS/IPS) | Detect abnormal TCP SYN traffic patterns and automatically generate alerts or block malicious connections. |
| Medium | Use DDoS Protection Services | Filter and absorb high volumes of malicious traffic before it reaches the organization's infrastructure, improving service availability during attacks. |
| Supporting | Deploy a Web Application Firewall (WAF) | Provide an additional security layer for HTTP/HTTPS traffic and complement network-level protections against web-based threats. |
| Supporting | Implement Continuous Network Monitoring | Continuously monitor network activity to detect anomalies, support incident response, and identify future attacks at an early stage. |

## 9. Skills Demonstrated

- Wireshark Analysis
- TCP/IP Fundamentals
- Incident Response
- Packet Inspection
- Network Security
- Technical Documentation

## 10. Conclusion

The investigation successfully identified a TCP SYN Flood attack as the root cause of the web server service disruption. Through the analysis of Wireshark packet captures, it was possible to recognize the abnormal TCP SYN traffic pattern, determine its impact on server availability, and understand how the accumulation of half-open TCP connections exhausted system resources.

This investigation also demonstrated the importance of network traffic analysis, incident response procedures, and timely mitigation strategies in reducing the impact of denial-of-service attacks. The recommended security controls—including SYN Cookies, Rate Limiting, IDS/IPS deployment, DDoS protection services, and continuous network monitoring—can significantly improve the organization's resilience against similar attacks in the future.